

VibeGuard: Mechanical Workshop & Carpentry Job Requisition

Component: Rig Baseplate (G4-1) & Mechanical Mounting Preparation

Institution: Jyothi Engineering College (Autonomous), Cheruthuruthy, Kerala

Department: Computer Science & Engineering / Electronics & Communication Engineering

Target Facility: Mechanical Workshop (Carpentry & Fitting Section)

Primary Contact: Amith Krishna Das (Hardware & Mechanical Lead)

1. Work Order Overview for Workshop Superintendent / Technician

Note:

Engineering Purpose: VibeGuard is an embedded edge vibration monitor. We are building a benchtop machine fault testbed using a 12V N20 gear motor that spins an off-center unbalance weight.

Physical Requirement: The motor and sensor must be anchored to a **stiff, dense, flat plate** so mechanical vibration transfers efficiently through the solid material into the ADXL345 sensor without flexing, warping, or acoustic resonance.

2. Material Options (Scrap Bin Offcut is Acceptable)

Please provide and cut one piece from workshop scrap stock matching either of these specifications:

Preference	Material Type	Dimensions ($L \times W \times T$)	Notes
Option 1 (Best)	Hard Plywood or Marine Plywood	$150 \times 150 \text{ mm}$ ($\approx 6 \times 6 \text{ inches}$), 10 mm to 12 mm thick	High stiffness, dampens tabletop rattle, holds wood screws firmly.
Option 2	Dense Hardwood Plank (Teak / Mahogany / Rubberwood)	$150 \times 150 \text{ mm}$, 12 mm to 15 mm thick	Must be planed flat on both faces. Zero warping.
Option 3	Clear Cast Acrylic Sheet (PMMA)	$150 \times 150 \text{ mm}$, 8 mm to 10 mm thick	Excellent aesthetics for exhibition, rigid transmission.

Caution:

Prohibited Materials: Do NOT use soft cardboard, styrofoam, low-density packing wood, or thin sheet metal ($\leq 3 \text{ mm}$). Thin metal flexes like a drumhead (membrane resonance), creating spurious harmonics.

3. Baseplate Layout & Drilling / Marking Template

+-----+			
	[0] 4 mm Mounting Hole (Corner 1)	Corner 2 [0]	
	+-----+		
	BREADBOARD & ESP32		
	ELECTRONICS ZONE		

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|               | (85 mm x 55 mm)               |
|               +-----+               |
|               |               |               |
|      N20 MOTOR POSITION              ADXL345 SENSOR      |
|      +-----+              +-----+              |
|      | [0] (17 mm) [0] |              | [0] (15) [0] |      |
|      | Motor U-Bracket | <--- 20 mm --> | Sensor Mount |      |
|      +-----+              +-----+              |
|               |               |               |
|               v               |               |
|      Rotating Cam Area              |               |
|               |               |               |
|      [0] Corner 3              Corner 4 [0]              |
+-----+
|<----- 150 mm ----->|

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Drilling & Machining Operations Requested:

- **Cutting & Squaring:**

* Cut baseplate stock to square dimensions: **$150\text{ mm} \times 150\text{ mm} \times 2\text{ mm}$** .

* Lightly sand/deburr all 4 edges and corners to prevent splinters.

- **Motor Bracket Pilot Holes (2 Holes):**

* **Location:** Center-left quadrant of the board.

* **Center-to-Center Spacing:** **17.0 mm** (matches the pre-punched holes in the aluminum N20 U-bracket).

* **Pilot Drill Bit:** **2.0 mm or 2.5 mm drill bit** to a depth of **8 mm** (prevents wood from splitting when tightening M3 wood screws).

- **Sensor Mount Pilot Holes (2 Holes):**

* **Location:** Exactly **20.0 mm parallel distance** from the motor bracket.

* **Center-to-Center Spacing:** **15.0 mm** (or **16.0 mm** for 3D bracket base).

* **Pilot Drill Bit:** $\$2.0\text{ mm}$ drill bit to a depth of $\$8\text{ mm}$.

- **Base Isolation Feet (Optional, 4 Holes):**

* Four $\$3.5\text{ mm}$ holes, 10 mm inset from each corner, to attach small rubber feet or standoffs so the rig does not vibrate across the table during tests.

4. Requisition Dialogue Script for Students

Whom to Approach: Carpentry Workshop Instructor or Fitting/Machine Shop Superintendent.

"Good morning Sir,

We are final-year students working on our microcontroller semester project (VibeGuard Vibration Fault Diagnostic Rig).

We need a small, sturdy baseboard from the workshop scrap bin to mount our test motor and vibration sensor:

*1. Could you help us cut a scrap offcut piece of **plywood or dense wood, roughly 15 cm by 15 cm, 10 to 12 mm thick?***

2. Could you help us drill two small 2 mm pilot holes with 17 mm spacing so we can screw down our miniature motor bracket?

*3. Do you also have **2 or 4 small self-tapping wood screws (around 10 mm length)** to fasten the bracket into the wood?*

Here is the dimension drawing. Thank you very much, Sir!"

5. Acceptance Criteria Checklist (Before Leaving Workshop)

- ☐ **Flatness:** Baseplate rests flat on a granite surface plate or workshop table with zero rock/wobble.
- ☐ **Dimensions:** Length and width are between $\$140\text{ mm}$ and $\$160\text{ mm}$.
- ☐ **Hole Spacing:** Bracket mounting holes align cleanly with the aluminum N20 U-bracket slots.
- ☐ **Screws Included:** 2 to 4 small wood screws received and tested for grip into the pilot holes.